

Ada Lovelace: A Woman Who Imagined the Future

Ada Lovelace was a mathematician from England who lived in the 1800s. At a time when women had fewer opportunities to study science and mathematics, Ada became fascinated by **numbers, machines and how they worked**.

She worked with Charles Babbage, who designed an early mechanical computer. Ada wrote a set of instructions for his machine that is often considered the **first computer program**.

Ada also imagined that computers could one day do much more than calculations. She thought machines might work with music, pictures and other kinds of information—ideas that were far ahead of her time.

Ada's story teaches us that **being curious and imagining new possibilities can lead to amazing ideas**. You don't have to know exactly what the future will look like to help create it.

Tebello Nyokong: A Scientist Who Never Gave Up

Professor Tebello Nyokong is a famous chemist who works at Rhodes University in South Africa. She grew up in a rural area and faced many challenges while going to school. When she was younger, some people told her that science was too difficult for girls. At first, she studied arts, but she later decided to follow her interest in mathematics and science.

Through hard work and determination, she became a leading scientist. Today, she researches ways to use light and special chemicals to help detect and treat cancer.

Tebello's story teaches us that where you come from does not decide what you can achieve. It is important to stay curious, believe in yourself and keep trying—even when something feels difficult.

Chevarra Hansraj: Exploring the Universe with Maths

Dr **Chevarra Hansraj** is a South African mathematician and scientist from **Durban, KwaZulu-Natal**. She became fascinated by space when she read about **black holes, neutron stars and galaxies** as a child.

She went on to study mathematics and physics and now researches **black holes, gravity and the universe** using mathematics. Her work helps scientists understand what happens around some of the most extreme objects in space.

One interesting thing about Chevarra is that **geometry was her least favourite topic in school mathematics**. Today, geometry is an important part of her work!

Chevarra's story teaches us that **you don't have to love every part of a subject straight away**. Keep asking questions, stay curious and give yourself time to discover what excites you.

Thifhelimbilu Daphney Bucher: From Rural Limpopo to Nuclear Physics

Dr **Thifhelimbilu Daphney Bucher** is a South African nuclear physicist who grew up in **rural Limpopo**. Her family did not have much money, but her mother worked hard to help her get an education.

Daphney became fascinated by physics and worked hard to study it. She went on to earn a **PhD in experimental nuclear physics** and became a researcher at South Africa's iThemba LABS, where scientists study the tiny particles that make up atomic nuclei.

Today, she continues to research **how atomic nuclei behave** and teaches the next generation of students.

Daphney's story teaches us that **your circumstances do not have to decide your future**. With curiosity, hard work and support, you can follow your interests and discover how far they can take you.

Seymour Papert: A South African Who Changed How Children Learn

Seymour Papert was a mathematician and computer scientist who was born in **Pretoria, South Africa**. He became fascinated by mathematics and computers and believed that children should learn by **exploring, experimenting and making things**.

Papert helped create **Logo**, a computer programming language that allowed children to give instructions to a small turtle on the screen. By using simple commands, children could create shapes, pictures and patterns while learning mathematics and programming.

His ideas helped change the way people think about **children, computers and learning**. He believed that making mistakes and trying again were important parts of discovering new ideas.

Seymour's story teaches us that **you don't just have to use technology—you can create with it**. Curiosity, creativity and trying things out can help you become a better problem solver.

Katherine Johnson: The Mathematician Who Helped Reach Space

Katherine Johnson was an American mathematician who loved **numbers and solving problems** from a young age. At a time when opportunities for Black women were limited, she worked hard to become a mathematician at **NASA**.

Katherine used mathematics to calculate the paths of spacecraft travelling through space. Her calculations helped astronauts travel safely, including **John Glenn's historic trip around Earth**.

She was known for being very careful with her work and never being afraid to **ask questions and check the answers**. Her mathematics helped NASA make important decisions about space missions.

Katherine's story teaches us that **maths can be much bigger than a classroom**. With curiosity, determination and careful thinking, the skills you learn in maths can help solve some of the world's biggest problems.

Maggie Aphane: A Mathematician Breaking Barriers

Professor Maggie Aphane is a South African mathematician who studies **topology**, a part of mathematics that explores shapes, spaces and how they are connected. She earned her degrees in mathematics from the **University of South Africa (UNISA)**, including her PhD in 2016.

Maggie became the **first South African woman and second woman overall to earn a PhD in Mathematics from UNISA**. She has also spoken about sometimes being the only Black woman at international mathematics conferences.

Today, she is a mathematics professor and Head of the Department of Mathematics and Applied Mathematics at **Sefako Makgatho Health Sciences University**. Her research connects mathematics with areas such as data science, networks and medical imaging.

Maggie's story teaches us: You don't have to see many people like yourself in a subject to belong there. **Keep learning, ask questions and make your own path in maths.**

Allan McLeod Cormack: The Mathematician Who Helped Us See Inside the Body

Allan McLeod Cormack was a **South African physicist and mathematician** born in Johannesburg. He became interested in using mathematics and physics to understand how things work.

Cormack developed mathematical ideas that showed how **X-rays taken from different angles could be used to create pictures of what is inside the body**. These ideas became an important part of the science behind **CT scans**.

In 1979, Cormack and British engineer Godfrey Hounsfield were awarded the **Nobel Prize in Physiology or Medicine** for their work in developing computer-assisted tomography.

Cormack's story teaches us: Mathematics is not only about numbers on a page. **Maths can help us understand the world, solve difficult problems and even improve people's health.**

Marie Curie: The Scientist Who Changed Our Understanding of Atoms

Marie Curie was a scientist born in **Poland** who later moved to France to study and do research. She loved learning about mathematics and physics, even when women had fewer opportunities to study science.

Marie and her husband, Pierre Curie, researched **radioactivity**, a term Marie helped develop. She discovered two new elements, **polonium and radium**, and studied how radiation could be used in science and medicine.

Marie became the **first woman to win a Nobel Prize** and the first person to win Nobel Prizes in two different scientific fields: Physics and Chemistry.

Marie's story teaches us: You can make important discoveries by being **curious, patient and willing to keep exploring difficult questions**. Your curiosity can take you further than you imagine.